#A The algorithm the key is to be used for.
#B The base64-encoded bytes of the key itself.

The JWE specification also includes an `enc` header that specifies the cipher used to encrypt the JSON body. This header is less error-prone than the `alg` header, but you should still validate that it contains a sensible value. Encrypted JWTs are discussed in section 6.3.3.

SPECIFYING THE KEY IN THE HEADER

To allow implementations to periodically change the key that they use to authenticate JWTs, in a process known as *key rotation*, the JOSE specifications include several ways to indicate which key was used. This allows the recipient to quickly find the right key to verify the token, without having to try each key in turn. The JOSE specs include one safe way to do this (the `kid` header) and two potentially dangerous alternatives listed in table 6.2.

> **DEFINITION *Key rotation*** is the process of periodically changing the keys used to protect messages and tokens. Changing the key regularly ensures that the usage limits for a key are never reached and if any one key is compromised then it is soon replaced, limiting the time in which damage can be done.

Table 6.2 Indicating the key in a JOSE header

Header	Contents	Safe?	Comments
`kid`	A key ID	Yes	As the key ID is just a string identifier, it can be safely looked up in a server-side set of keys.